



Aplikasi Mobile Sistem Informasi Akademik Mahasiswa

Berbasis Flutter + Firebase + Fitur PDF & Notifikasi Email



Tujuan Praktik

Mahasiswa mampu:

- Mengembangkan aplikasi mobile dengan Flutter
 - Mengelola data akademik menggunakan Firebase (Firestore & Auth)
 - Membuat laporan PDF (KHS/KRS)
 - Mengirim notifikasi email otomatis
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Fitur Aplikasi

1. Login / Register Mahasiswa
 2. Dashboard (profil + ringkasan akademik)
 3. Data Mata Kuliah & Nilai
 4. KRS (pengambilan mata kuliah)
 5. KHS (hasil studi)
 6. Generate laporan PDF
 7. Kirim notifikasi email (misalnya saat nilai keluar)
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Teknologi yang Digunakan

- Flutter (UI & logic)
 - Firebase Authentication (login)
 - Cloud Firestore (database)
 - Firebase Cloud Functions (opsional untuk email)
 - Plugin:
 - pdf
 - printing
 - mailer / firebase_functions
-



Struktur Praktik

♦ 1. Setup Project

```
flutter create siakad_app
```

```
cd siakad_app
```

Tambahkan dependency di `pubspec.yaml`:

```
firebase_core: ^latest

firebase_auth: ^latest

cloud_firestore: ^latest

pdf: ^latest

printing: ^latest
```

♦ 2. Desain Database (Firestore)

Collection:

- `users`
- `courses`
- `krs`
- `grades`

Contoh struktur:

```
users:

  - uid

  - nama

  - email


grades:

  - uid

  - matkul

  - nilai

  - semester
```

♦ 3. Fitur Login

Gunakan Firebase Auth:

```
await FirebaseAuth.instance.signInWithEmailAndPassword(  
  
  email: email,  
  
  password: password,  
  
);
```

♦ 4. Menampilkan Data Nilai

```
StreamBuilder(  
  
  stream: FirebaseFirestore.instance.collection('grades').snapshots(),  
  
  builder: (context, snapshot) {  
  
    if (!snapshot.hasData) return CircularProgressIndicator();  
  
    return ListView(  
  
      children: snapshot.data!.docs.map((doc) {  
  
        return ListTile(  
  
          title: Text(doc['matkul']),  
  
          subtitle: Text("Nilai: ${doc['nilai']}"),  
  
        );  
  
      }).toList(),  
  
    );  
  
  },  
);
```

```
);
```

◆ 5. Generate Laporan PDF (KHS)

```
import 'package:pdf/widgets.dart' as pw;

final pdf = pw.Document();

pdf.addPage(

  pw.Page(

    build: (pw.Context context) {

      return pw.Column(

        children: [

          pw.Text("Kartu Hasil Studi"),

          pw.Text("Nama: Gunawan"),

          pw.Text("Matkul: Pemrograman Mobile - A"),

        ],

      );

    },

  ),

);
```

Tampilkan / print:

```
await Printing.layoutPdf(  
  
    onLayout: (format) async => pdf.save(),  
  
);
```

♦ 6. Notifikasi Email

Opsi sederhana (client side):

Gunakan package mailer:

```
final smtpServer = gmail('email@gmail.com', 'password');  
  
final message = Message()  
  
  ..from = Address('email@gmail.com')  
  
  ..recipients.add('tujuan@gmail.com')  
  
  ..subject = 'Nilai Baru'  
  
  ..text = 'Nilai kamu sudah keluar!';  
  
await send(message, smtpServer);
```

Opsi lebih aman (disarankan):

- Gunakan Firebase Cloud Functions
- Trigger saat data nilai ditambahkan

Tugas Praktik

1. Tambahkan fitur:
 - Filter nilai per semester
2. Export PDF berisi:
 - Nama mahasiswa
 - Daftar nilai lengkap
3. Kirim email otomatis saat nilai ditambahkan

 Penilaian

- UI/UX (20%)
- Integrasi Firebase (30%)
- Fitur PDF (20%)
- Notifikasi Email (20%)
- Kerapihan kode (10%)